



CHILDHOOD CANCER CLINICAL DATA COMMONS (C3DC) User Guide

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Introduction and Overview

The Childhood Cancer Clinical Data Commons (C3DC) is an open-access resource that helps researchers find and use deidentified, participant-level clinical data from childhood cancer studies. The data are harmonized using standardized terms and a [common data model](#), making it easier to compare and analyze data across studies.

The [C3DC Home Page https://clinicalcommons.ccdi.cancer.gov/home](https://clinicalcommons.ccdi.cancer.gov/home) allows users to navigate to key sections of the application, including the Explore Dashboard (Explore Participants and Explore Files), Cohort Analyzer, Studies summary, Data Model, and About pages. At the bottom of the Home page, there are links to brief descriptions of key sections of the C3DC application.

Users can navigate to the Studies Page to view the list of studies in the application, including counts for participants, diagnoses, samples, and files. By clicking on a Study ID (e.g., phs000463), users can access detailed information about the study. Please follow this [link](#) for instructions on how to access the controlled access data.

The [data model](#) is developed collaboratively with multiple organizations to establish standard terms for pediatric cancer. In this harmonization effort, we are using CDEs (Common Data Elements) to enhance data accuracy, consistency, and interoperability across health research studies. CDEs are defined in the caDSR (Cancer Data Standards Registry and Repository) and provide controlled terms, vocabularies, detailed information on data representation, and robust metadata. The CCDI data model schema consists of well-defined classes with attributes and permissible values.

This document describes a high-level overview of the features of the C3DC Explore Dashboard. Investigators are encouraged to explore C3DC themselves or use this guide as a primer.

CCDI Explore Dashboard and Cart

The Explore Dashboard is the main interface for searching, visualizing, and exporting data. Users can apply faceted filters, view dynamic charts, and inspect participant- and file-level tables that update in real-time based on selected criteria. The Explore Dashboard enables researchers to find CCDI data within a single study or across multiple studies and create synthetic cohorts based on filtered search (i.e., demographics, diagnosis, samples, etc.). Researchers can explore the data in two ways: [Explore Participants](#), which provides a participant-centric view, and [Explore Files](#), which provides a file-centric view. Using filters in either view, researchers can review available open-access information through visual summaries, browse row-level data in tabs, and toggle between the two views to identify data sets relevant to their research questions.

After applying all relevant filters to reduce the search space to records of interest, users can then add desired files to the cart from the Explore Files view. From the cart, they can download a manifest for the selected data or take the manifest directly into the Cancer Genomics Cloud (CGC). To access controlled data, users must request them at the [controlled-access login page](#) on the database of Genotypes and Phenotypes ([dbGaP](#)) (see [Appendix A](#)). Note that bulk downloads are only possible through a Command Line Interface (CLI) client as described in [Appendix B](#).

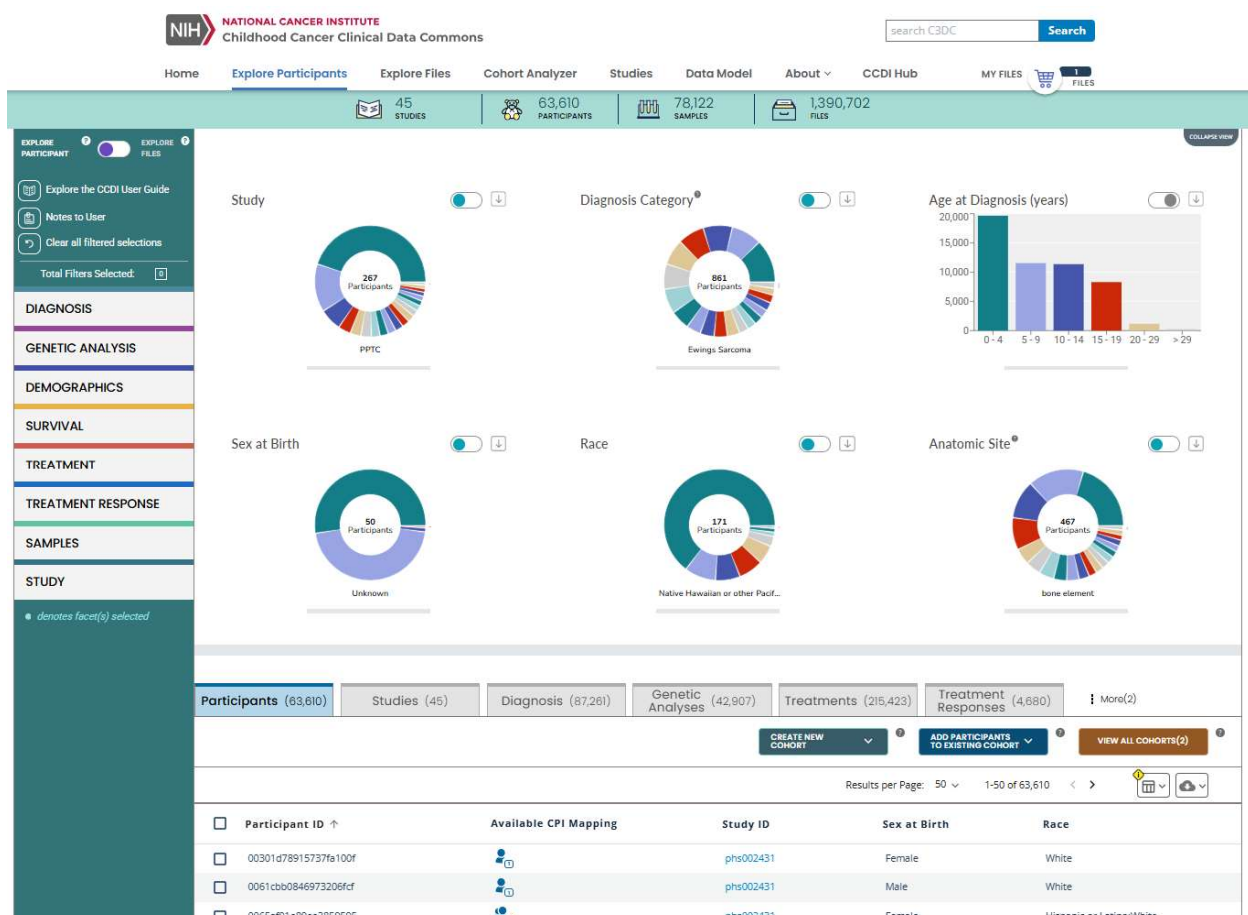


Figure 1: Explore Participants – Faceted Search (left), Visualization (top), and Data Tables (bottom)

Facet Search to find Participants and Files

The CCDI Explore Dashboard provides row-level metadata for CCDI study participants and their data objects for review with a filtered search, select visualizations, and an exportable table of results. Here's how to find and filter information on the Explore Dashboard:

1. C3DC is located at clinicalcommons.ccdi.cancer.gov. From the C3DC Home page, navigate to the Explore Dashboard by clicking "[Explore Participants](#)".
2. On the Explore Participants view, users can filter row-level data for participants and view them as chart visualizations or in tabular summaries. This view is participant-centric, meaning that filtering criteria and results return de-identified information about a participant and their related studies, collected samples, or created files. Click on a facet to view the filter criteria available for that category.
3. Search criteria are displayed in the left panel. Faceted filtering may be done by text searches, uploading a list of participant IDs, numerical sliders, or checkbox selections. Users can apply multiple filtering criteria at the same time in a search. Users can view and clear their current selection(s) in the query summary at the top of the widgets.
4. Filtering the search will update the Explore Dashboard's visualizations and the results tables. Each results table will be updated with information that meet the filtered criteria for the corresponding facet. Information displayed by default on each Explore Participants table is described below:
 - a. "Participants": Characteristics of a participant in the Explore Dashboard. Participants belong to a study, and they may have one or more samples, diagnoses, or files associated with them. Participant mappings from the CCDI Participant Index (CPI) have a summary of these mappings available from the table, and full mapping details are available by clicking on the adjacent icon.
 - b. "Studies": Studies that are a part of the Explore Dashboard. Participants, diagnosis, samples, and files all belong to a CCDI study.
 - c. "Diagnosis": Age, anatomic site, classification, and other attributes of each participant's diagnosis are included here.
 - d. "Genetic Analysis": Analysis ID, Gene Symbol(s) of implicated genes, and reported status and significance are part of the genetic analysis summary.
 - e. "Treatments": This tab includes treatment IDs, age at start and end of treatments, and treatment type and agent.
 - f. "Treatment Responses": This tab includes response ID, the response, and the age at which a response was recorded.
 - g. "Survival": This tab includes survival ID, last known survival status, and age at last known survival status.
 - h. "Samples": Samples available from participants within the Explore Dashboard. Samples belong to a participant and can be associated with one or more files.
5. Visible columns in each table can be customized by clicking the "View columns" button in the upper righthand corner of the table and selecting or deselecting available columns. Note that Participant ID, Sample ID, and Study ID cannot be removed from any table, and File Name cannot be removed from the Files table.

- Users can show, hide, and copy the URL used to construct their filtered view using the toggle (Show/Hide Query URL) and copy buttons at the top of the widgets.

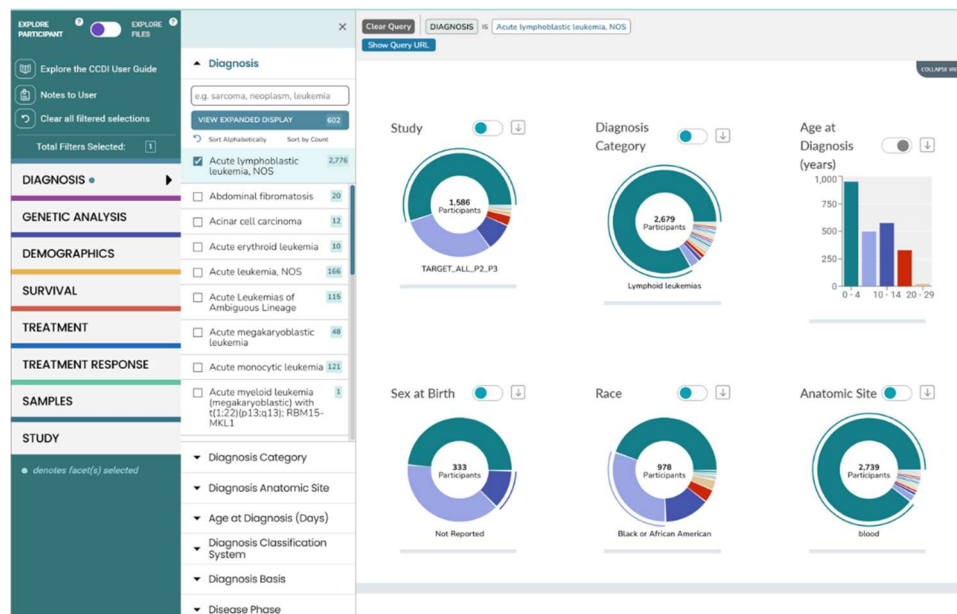


Figure 2: Performing a faceted search on the C3DC Explore Participants Page

Similarly, users can toggle to “[Explore Files](#)” and apply the same types of filters on the corresponding file data. Row-level data are limited to a single “**Files**” view, including various annotations to describe that given file. Files may belong to a study and may be associated with one or more participants or samples. Files may also be of many types, including sequencing, proteomics, imaging files, etc. DICOM imaging files are currently available for the Genomic Sequencing of Pediatric Rhabdomyosarcoma (phs000720) and Molecular Characterization Initiative (phs002790) studies and can be accessed directly from the Imaging Data Commons (IDC) Data Portal. File paths to images are provided in the downloadable study manifest described in the [Downloading Metadata from the Studies table](#) section. File attributes, including but not limited to file name, can be searched in this tab.

Chart Visualization

The **Stats Bar**, **Visualization Section**, and **Data Tables** dynamically update based on applied filters from both the Explore Participants and Explore Files views.

- Chart visualizations illustrate the proportions of each value in the form of either pie chart or histogram.
- Users can easily switch between pie chart and histogram through the toggle button by each visualization.
- Download button for each visualization is also available for users to download a PNG file.

Synonym Search

When searching for a participant, users can use any recognized synonym from the CCDI Participant Index (CPI). The system will identify the synonym, note that it matches a known participant, and display the corresponding participant_id and details in the results table below.

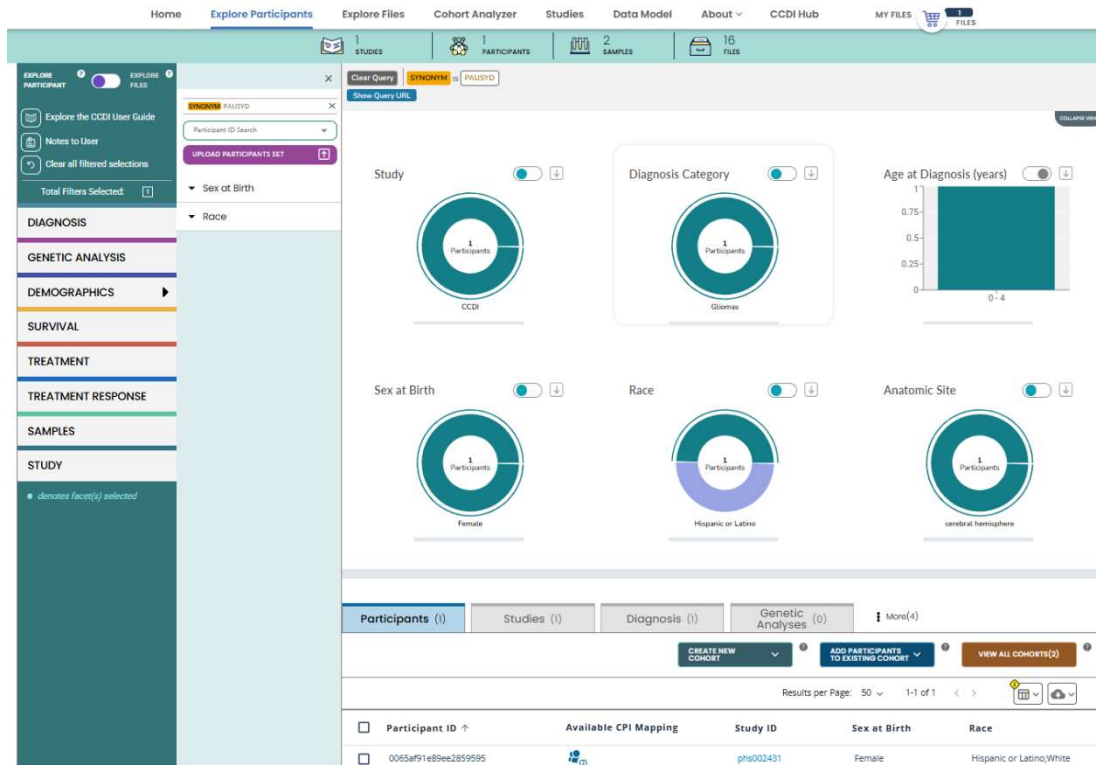


Figure 3: Searching by synonym capabilities and match confirmation

Download Harmonized Data

Users can download the contents of all tabs by selecting the "Download Data" button under the table tab headers. Users can download filtered data in either CSV or JSON formats.

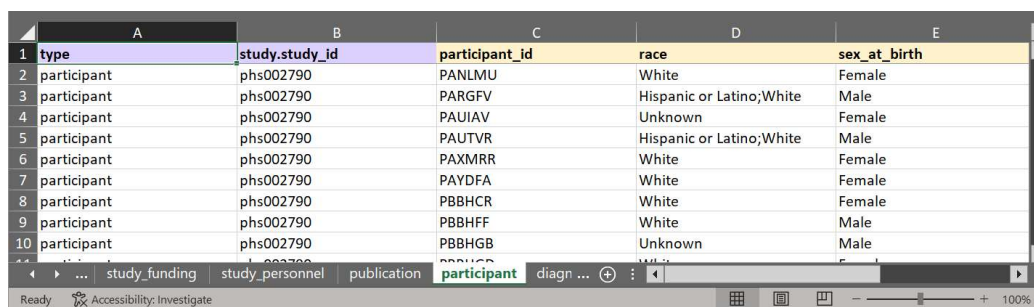
Participants (63,610)					
Studies (45)					
Diagnosis (87,261)					
Genetic Analyses (42,907)					
More(4)					
Results per Page: 50 1-50 of 63,610					
Download Data					
CSV JSON					
☐	Participant ID ↑	Available CPI Mapping	Study ID	Sex at Birth	Race
☐	00301d78915737fa100f		phs002431	Female	White
☐	0061cbb0846973206fcf		phs002431	Male	White
☐	0065af91e89ee2859595		phs002431	Female	Hispanic or Latino;White
☐	008b04a84717e7d007a4		phs002431	Male	White
☐	009f9efaedc9602a28fb		phs002431	Female	White
☐	00c5e4372375eb3627f3		phs002431	Female	White

Figure 4: Download button to download harmonized data in the Explore Page

Downloading Metadata from the Studies table

From the CCDI Explore Dashboard, users can download all open metadata for each study from the “Studies” tab to further filter data and build cohorts. For instance, additional filtering by diagnosis of interest can generate a set of participants, and the resulting manifest can be uploaded into the CGC. As an example, the following steps explain how to download the metadata for the CCDI Molecular Characterization Initiative:

1. Using the process described above, open the “STUDY” set of filters from the lefthand menu, expand the “Study Name” category, and scroll down to find “Molecular Characterization Initiative.”
2. Select the checkbox corresponding to “Molecular Characterization Initiative” and see the Dashboard reload, filtered for this study’s details.
3. Navigate to “Studies” in the results tables and locate the “Manifest” column.
4. Click the “Download study manifest” icon in the “Manifest” column to download the metadata for this study.
5. Open the resulting file on your local machine to browse the resulting metadata tables.



	A	B	C	D	E
1	type	study.study_id	participant_id	race	sex_at_birth
2	participant	phs002790	PANLMU	White	Female
3	participant	phs002790	PARGFV	Hispanic or Latino;White	Male
4	participant	phs002790	PAUIAV	Unknown	Female
5	participant	phs002790	PAUTVR	Hispanic or Latino;White	Male
6	participant	phs002790	PAXMRR	White	Female
7	participant	phs002790	PAYDFA	White	Female
8	participant	phs002790	PBBHCR	White	Female
9	participant	phs002790	PBBHFF	White	Male
10	participant	phs002790	PBBHGB	Unknown	Male

Figure 5: Study metadata export file browsable on local machine

Note the full study manifest can also be downloaded from the associated study details page accessible from the Studies list in the application menu. Links to the [CCDI cBioPortal Cancer Data Explorer](#) are also available from this page for studies with related data available through the CCDI cBioPortal.

[Appendix C](#) details the process for generating a DRS manifest from the downloaded study metadata tables to be compatible with the CGC.

Creating an Exportable File Manifest from the Cart

In addition to the study-specific downloads, users can also export each row-level metadata element based on their selections within the Explore Dashboard. Here’s how to create a manifest file of filtered information on the Explore Dashboard:

1. On the Files tab of the Explore Files view, users can select a row using the checkbox at the start of the row. Multiple rows can be selected within a table, even across pages of the table. Use the checkbox at the top of the checkbox column to select or deselect all rows.

2. After selecting desired rows, add files for that element to the My Files shopping cart by clicking either the “ADD ALL FILTERED FILES” or “ADD SELECTED FILES” button. Note that selection of items in each tab depends on the specific content of that tab.
3. To navigate to the shopping cart, select “MY FILES” or the shopping cart icon on the menu bar.



Figure 6: Menu bar with My Files shopping cart

4. The shopping cart feature enables users to select and manage files. It’s a simple way to keep track of data and files during a session. Selecting the “DOWNLOAD MANIFEST” button from the “AVAILABLE EXPORT OPTIONS” dropdown menu will produce a comma-separated values (CSV) file manifest of the items within the cart.

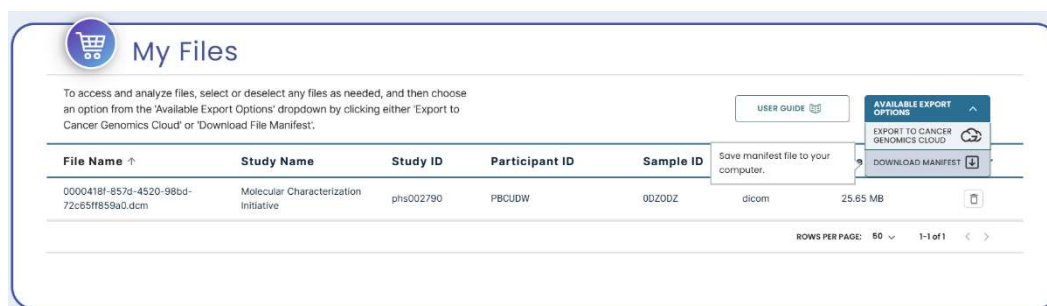


Figure 7: The Explore Dashboard Cart page with “Available Export Options” button

5. Users can then download this manifest file locally or upload it in the CGC ([Appendix C](#)). Similarly, users can instead select the “EXPORT TO CANCER GENOMICS CLOUD” button from the “AVAILABLE EXPORT OPTIONS” dropdown menu to load the resulting manifest directly into their CGC account.

Note that the Cart has a maximum capacity of 200,000 files, which may limit the ability to create very large manifests for use in the CGC. Should users need to create a manifest containing more than 200,000 files, they can either create manifests from the cart in batches (containing up to 200,000 files in each batch) or use the comprehensive metadata downloads from the Explore page “Studies” tab to create a manifest that can take all data for a given study into the CGC. Longer term solutions are being researched.

Cohort Selector

The Cohort Selector enables users to create a cohort with a size of up to 4,000 participants and manage up to 20 cohorts. This feature offers flexibility to researchers, allowing them to create cohort groups according to their specific requirements.

Participants (63,610)				
Studies (45)				
Diagnosis (87,261)				
Genetic Analyses (42,907)				
More(4)				
CREATE NEW COHORT ADD PARTICIPANTS TO EXISTING COHORT VIEW ALL COHORTS(2)				
All Participants Selected Participants				
Per Page: 50 1-50 of 63,610 < > 📄 ☁️				
☐ Participant ID ↑	Available CPI Mapping	Study ID	Sex at Birth	Race
☐ 00301d78915737fa100f		phs002431	Female	White
☐ 0061cbb0846973206fcf		phs002431	Male	White
☐ 0065af91e89ee2859595		phs002431	Female	Hispanic or Latino;White

Figure 8: Cohort Selection features visible on the Explore page Table tabs (see choice buttons in orange rectangle)

Users can do the following:

- Create New Cohort:
 - Users can select participant IDs from the table or choose to add all participants based on the faceted results and create a new cohort.
 - Users can name and describe the cohort for easy reference.
 - A user can add up to 4000 participants in each cohort.
- Add Participants to Existing Cohort:
 - Select participants and add them to existing cohorts or remove them. Entire cohorts can also be deleted as needed.
- View All Cohort(s): View a list of all created cohorts, making it easier to manage and analyze groups.
 - Cohort ID: Create your own IDs to identify saved cohorts
 - Cohort Description: Create descriptions for saved cohorts
 - Save Changes: Save the changes made to the selected cohort. This includes changes to cohort ID, cohort description, and any participants.
 - **Copy Cohort:** Create a copy of an existing cohort and add or remove participants as needed. This action creates a new cohort with the same participants and settings, with “Copy” appended to the cohort name.
 - Download Selected Cohort:
 - Download the metadata of selected cohort in one of two formats.
 - Manifest CSV: a list of participant IDs and high-level metadata.
 - Metadata JSON: a JSON file containing all metadata information for the participants in the selected cohort, including CPI synonyms
 - View Cohort Analyzer: Navigate to the Cohort Analyzer from the cohort list.
 - Explore in CCDI Hub: Export cohorts (up to 4000 participants for each cohort) that open the CCDI Hub with pre-filtered data based on selected participants.

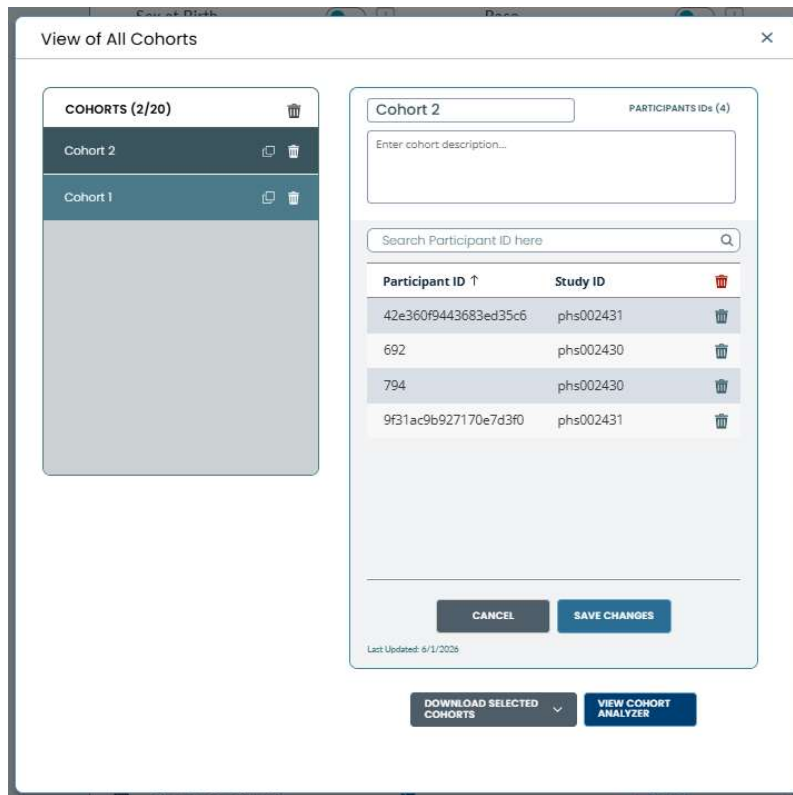


Figure 9: View All Cohorts popup allows users to manage up to 20 cohorts – users are given choices to download cohort metadata or view cohort analyzer

C3DC Cohort Analyzer

The Cohort Analyzer offers a powerful method to explore how various clinical attributes overlap and differ across multiple groups. The Cohort Analyzer is designed to compare up to three cohorts and visualize their intersections through an interactive Venn diagram, corresponding histograms, survival analysis visualizations, and a data table. This feature leverages cohorts created on the Explore Participants page, enabling users to analyze key relationships and distinctions based between datasets effectively. By visualizing the shared and unique data points using a Venn diagram, users can identify common patterns or variations in key clinical variables such as diagnosis, treatment, and participant characteristics.

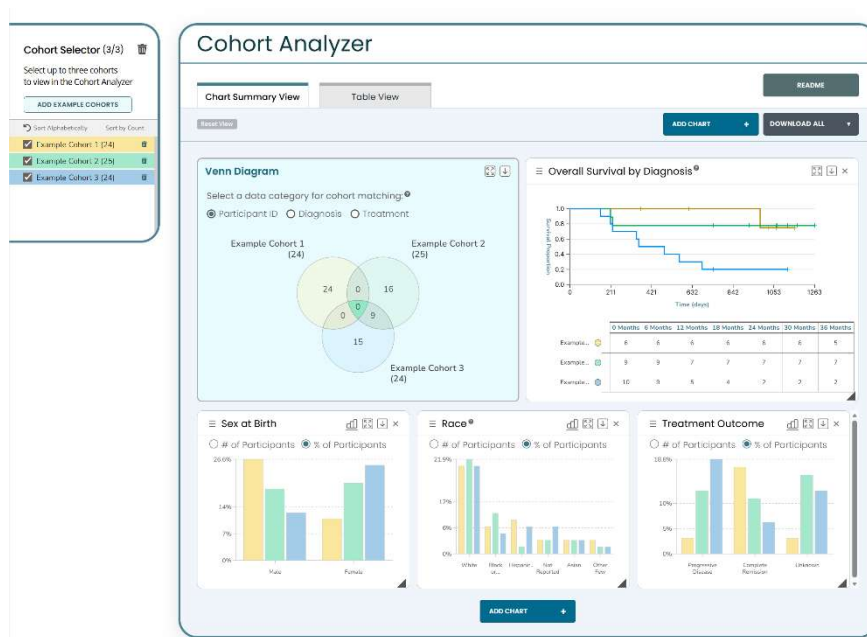


Figure 10: Cohort Analyzer landing page

Customizable Properties

The radio buttons allow users to select different properties for comparison. The Venn diagram of Participant ID shows the number of participants shared between different sets, while the Venn diagrams of Diagnosis or Treatment display the number of unique values under each category.

Enhanced Analytical Capabilities

Users will be able to visualize overlaps and unique attributes within each cohort. In addition, users can:

- Investigate specific sections of the Venn diagram to view participant-level details from the corresponding table view.
- Export results, including the data table, histograms, and Venn diagram, for further analysis or integration into other platforms.

- Use advanced filters to refine cohort comparisons, such as narrowing by treatment or specific diagnosis.
- Download results: The cohort results can be downloadable as a CSV with individual high-level metadata or a JSON file with comprehensive metadata, including CPI synonyms.
- Build in Explore Dashboard: Export analyses into a pre-filtered view within the Explore Dashboard for streamlined review and exploration.
- “Add Example Cohorts” button allows user to explore cohort analyzer features easily by adding 3 mock cohorts.

Cohort Analyzer Tutorial

To start using the Cohort Analyzer, users will first need to select the cohorts they want to analyze. As cohorts are added, the system will automatically keep track of the cohorts on the left side Cohort Selector.

Select the first cohort by clicking the checkbox in the Cohort Selector sidebar. The Venn diagram and table update to display cohort information based on the selected radio button (Participant ID, Diagnosis, or Treatment). By default, histograms for Sex at Birth, Race, and Treatment Outcome, as well as a Kaplan–Meier survival plot with accompanying risk table, are displayed. Users can also click the Add Chart to show an additional histogram for Treatment Type. The corresponding Table View shows the list of participants in the selected Cohort.

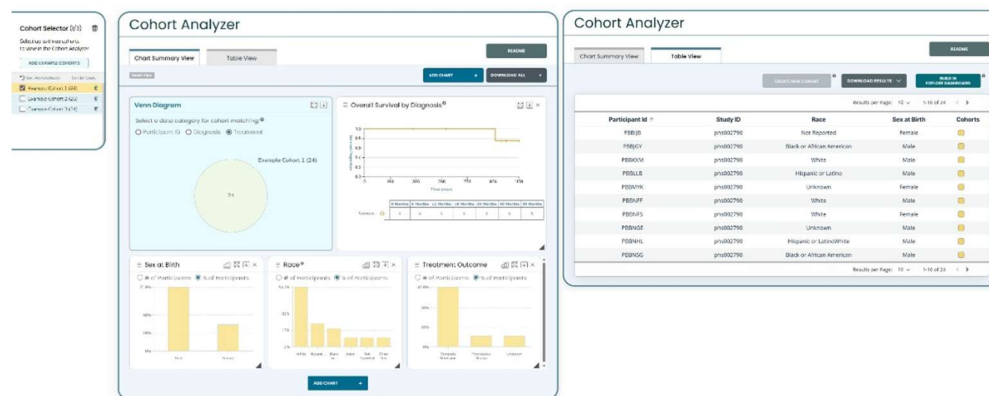


Figure 11: One Cohort Selected

Select another selector in the Cohort Selector to see the results update. This time, if there are common participants between both cohorts, the diagram will show the shared participants in the intersection between the two. Clicking the Diagnosis radio button shows a Venn diagram of unique and shared diagnosis values between two cohorts. In the Table View with none of the Venn diagram selected, it will display all participants and their respective cohort. Selecting part of the Venn diagram will update the table content accordingly. With a section selected, a user can also create an entirely new cohort with these filtered participants by clicking the “Create New Cohort” button in the Table View.

Note that the number in parentheses by the cohort's name in the Venn diagram represents the count of unique records for that radio button selection. The number inside the Venn diagram sections are the count of unique values for that radio button selection. Finally, the count next to a cohort in the Cohort Selection side bar indicates the total participants in the cohort.

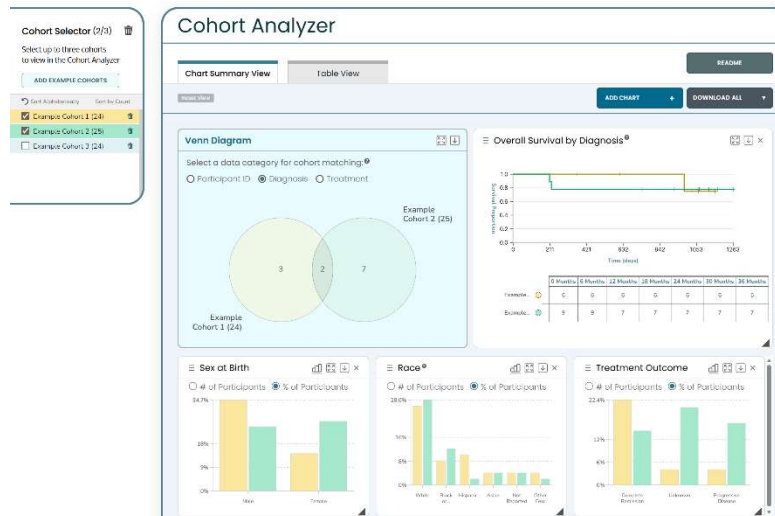


Figure 12: Two Cohorts Selected

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Appendix A: Submitting a Data Access Request to the Database of Genotypes and Phenotypes (dbGaP)

CCDI studies with controlled-access data are registered with the National Center for Biotechnology Information's database of Genotypes and Phenotypes (dbGaP), which maintains a list of the studies' subject IDs, sample IDs, and consents.

Eligible investigators interested in obtaining a controlled data set should watch the instructional video on applying for controlled access data and consult the Tips on Preparing a Successful Data Access Request. A step-by-step breakdown of the data access request process is located in this [guide](#).

Appendix B: NCI Data Commons Framework Services (DCFS): Controlled Data Access Instructions

NCI Data Commons Framework Services (DCFS), powered by Gen3, facilitates data authorization in a secure and scalable manner. DCFS's Indexd service provides permanent digital IDs for data objects. These IDs can be used to retrieve the data or query the metadata associated with the object.

CCDI data is available for download using the DCFS and bulk downloads are only possible using this method. To gain access to controlled data, researchers must first have an NIH eRA Commons account for authentication, after which they will need to obtain authorization (via an active DCFS login account) to access the data in dbGaP.

Below are instructions for using the Data Commons Framework (DCF) user interface or the DCF Gen3-client to access CCDI data.

File Download Procedure via User Interface

To download a study-specific research data distribution file with the DCF Services Portal interface, a researcher must execute the following steps:

1. Please navigate to nci-crdc.datacommons.io and click the “RAS Login” button. Login to the NCI DCF Services portal at nci-crdc.datacommons.io/login (Figure B1).

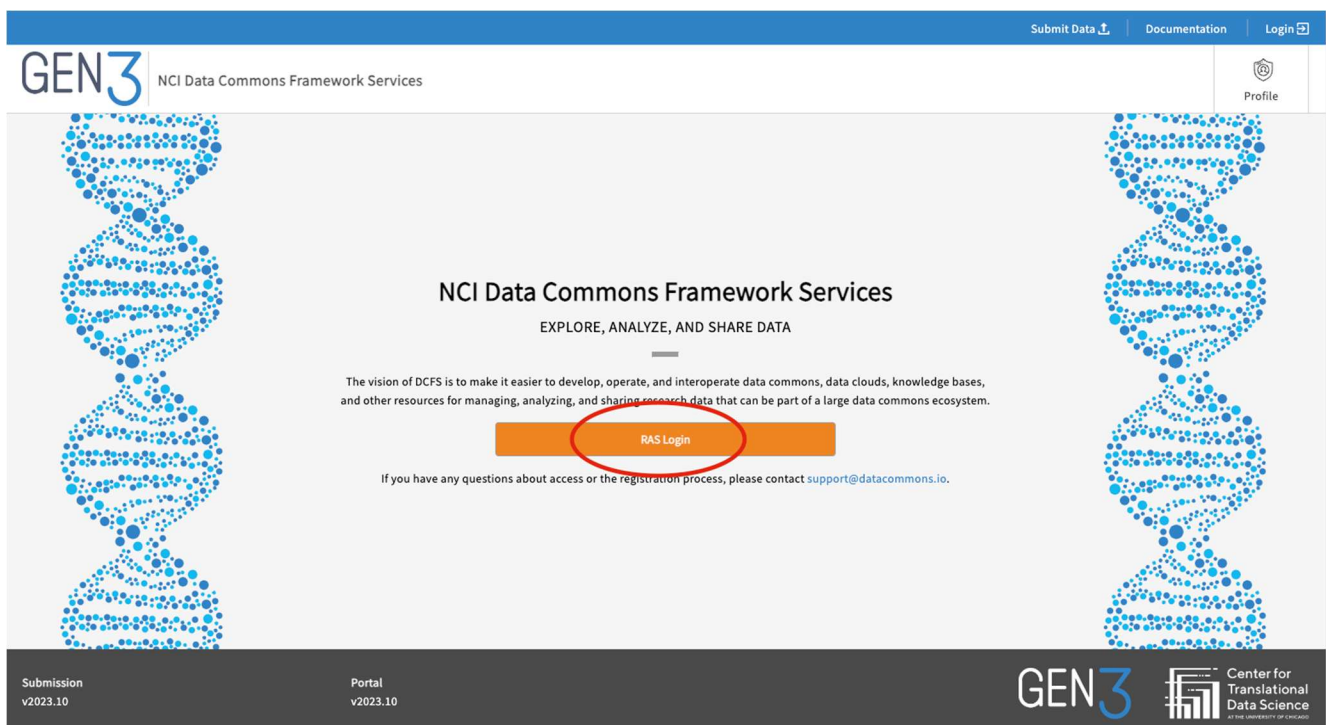


Figure B1: NIH DCF homepage with the NIH Researcher Auth Service (RAS) login highlighted

- Once logged in, click the “Profile” section in the top right corner and review your project access to confirm study access (Figure B2).

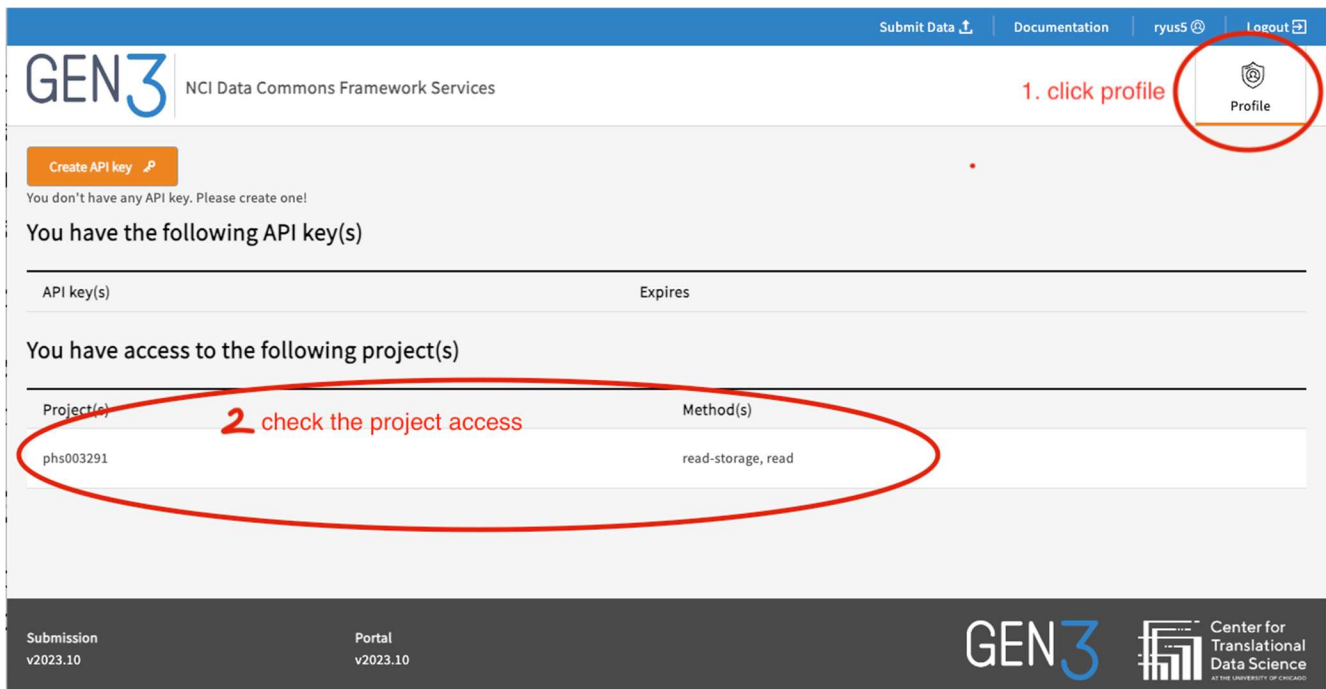


Figure B2: DCF Profile page highlighting “Profile” and the accessible projects.

- Add the file GUID after the final backslash in this URL: <https://nci-crdc.datacommons.io/user/data/download/>. Paste the URL you created in a browser address field and press Enter or Return.
- The NCI DCF Services Portal will respond by providing a JSON document with a new (signed) URL for the requested data file. Copy the signed URL.
- Paste this new signed URL into the browser address field and press Enter or Return (Figure B3).
 - Note: Once issued, the signed URL provided is valid for a relatively short period of time.
- The NCI DCF Services Portal will respond by displaying a URL. Click the URL to download the file (Figure B3).

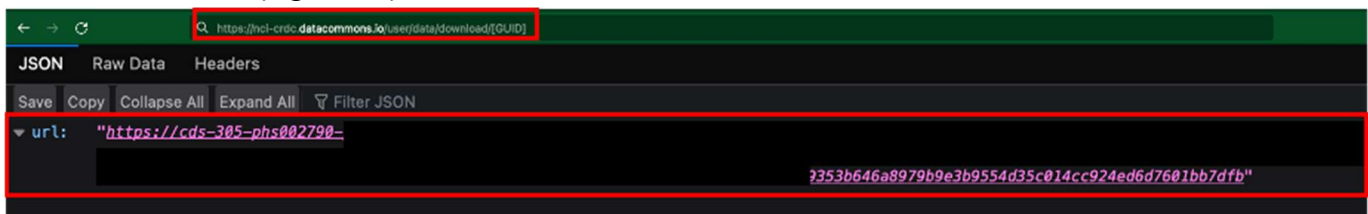


Figure B3: DCF Service Portal displaying the signed access URL and the file download URL

Note: If errors or problems are experienced during the file downloading process above, please contact the CCDI mailbox for assistance.

File Download Procedure via Command Line Interface (CLI) client

To download a study-specific research data distribution file with a CLI client, a researcher must execute the following steps:

1. Obtain the Gen3-client command-line tool from GitHub.
2. Install and configure the client based on the Gen3 instructions.
 - a. These instructions include signing into the DCF web client and obtaining a downloaded JSON API key from the Profile page, and then configuring the client.
 - b. The API endpoint that will be used for DCF configuration is 'https://nci-crdoc.datacommons.io'.
3. Obtain either a GUID or manifest of GUIDs for the data files of interest from the CCDI Explore page or the Explore Dashboard exportable manifest.
4. Create a Gen3 structured manifest:

```
[
  {
    "object_id": "dg.4DFC/{guid_1}"
  },
  {
    "object_id": "dg.4DFC/{guid_2}"
  },
  ...
  {
    "object_id": "dg.4DFC/{guid_n}"
  }
]
```

5. Download the file(s) using the Gen3 client (either the single or multiple download option).

For more information on this process, please visit the Gen3 documentation page.

Appendix C: The Cancer Genomics Cloud (CGC)

The Seven Bridges Cancer Genomics Cloud (CGC), powered by Velsera and funded by NCI, is a flexible cloud platform that enables analysis, storage, and computation of large cancer data sets. The CGC provides a user-friendly portal to access and analyze cancer data without having to learn how to program.

Creating a CGC Account

You need an account to access and analyze CCDI data on the CGC platform. Note that a list of all CCDI studies released is also available. The following instructions describe the process to create a CGC account.

1. From the CGC home page, cancer-genomics-cloud.org, click “REGISTER” or “LAUNCH” in the center of the page (Figure C1).

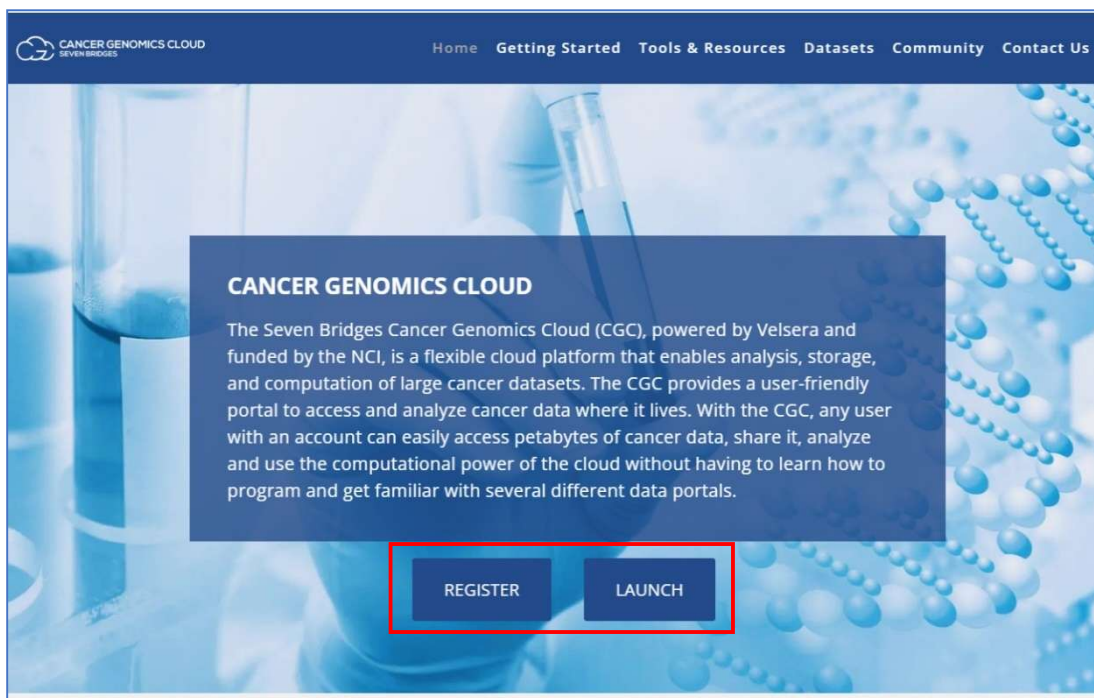


Figure C1: CGC home page with “REGISTER” and “LAUNCH” buttons highlighted in a red box.

2. On the login screen, click on “Log in with eRA Commons” (Figure C2).

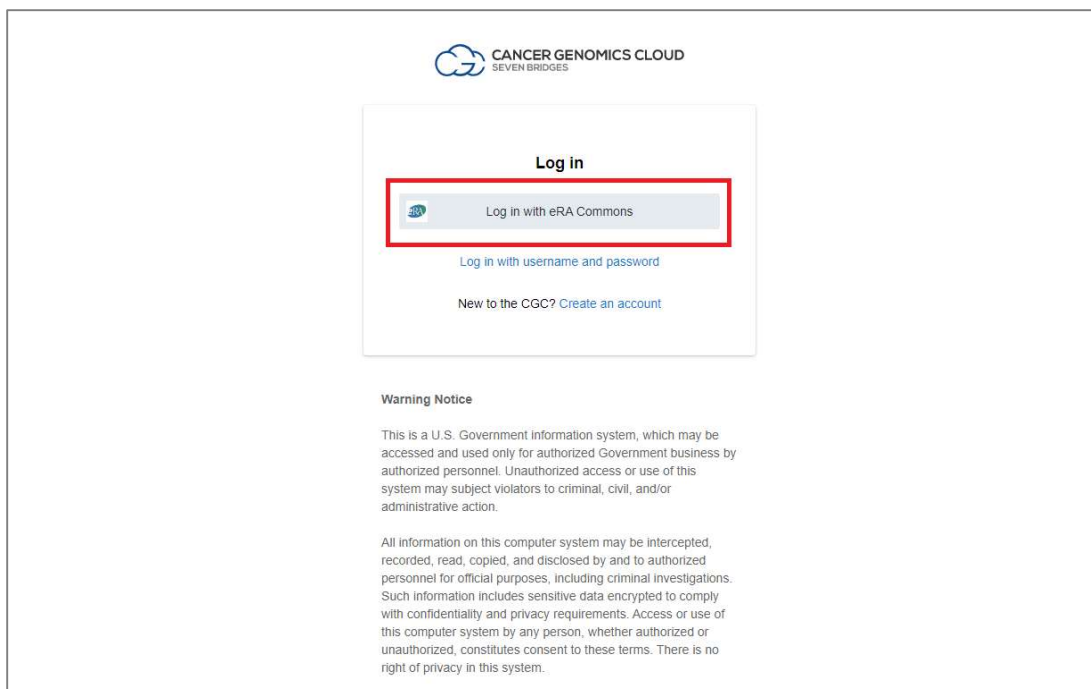


Figure C2: eRA Commons login screen for the CGC, with login button highlighted in a red box

3. On the login screen, enter the eRA Commons account credentials associated with your approved dbGaP study and click “Sign In” (Figure C3).

Please note that if you receive an error message when logging in here, you can confirm that your eRA Commons username and password are correct by logging in to the eRA Commons site.

For additional details about ensuring your Velsera eRA Commons account is correct and correctly configuring the download restriction and network access toggle at the project level, please refer to the following documentation:

- docs.cancergenomicscloud.org/docs/modify-project-settings
- docs.cancergenomicscloud.org/docs/define-download-restriction

If you’ve previously logged into the eRA Commons site, you may need to clear your web browser’s cache or use “incognito” mode to ignore cached data and cookies so you can enter and test your credentials. If you receive an error message on that site as well, you may need to reset your eRA Commons password.

Figure C3: Login page for CGC, with username and password credentials sections highlighted in a red box.

4. If you agree to the “Consent to Share Information” on the following page, click the “Grant” button to continue (Figure C4).

Figure C4: Consent to share information page with red box highlighting the “Grant” button to confirm consent to share information with the CGC

5. If you agree to authorize Gen3 DCFS to share your account and authorization information to access the data sets for which you have been approved, click the “Yes, I authorize” button (Figure C5). Note that Gen3 is an authorization system that uses eRA Commons as an authentication tool and allows access to the CGC system.

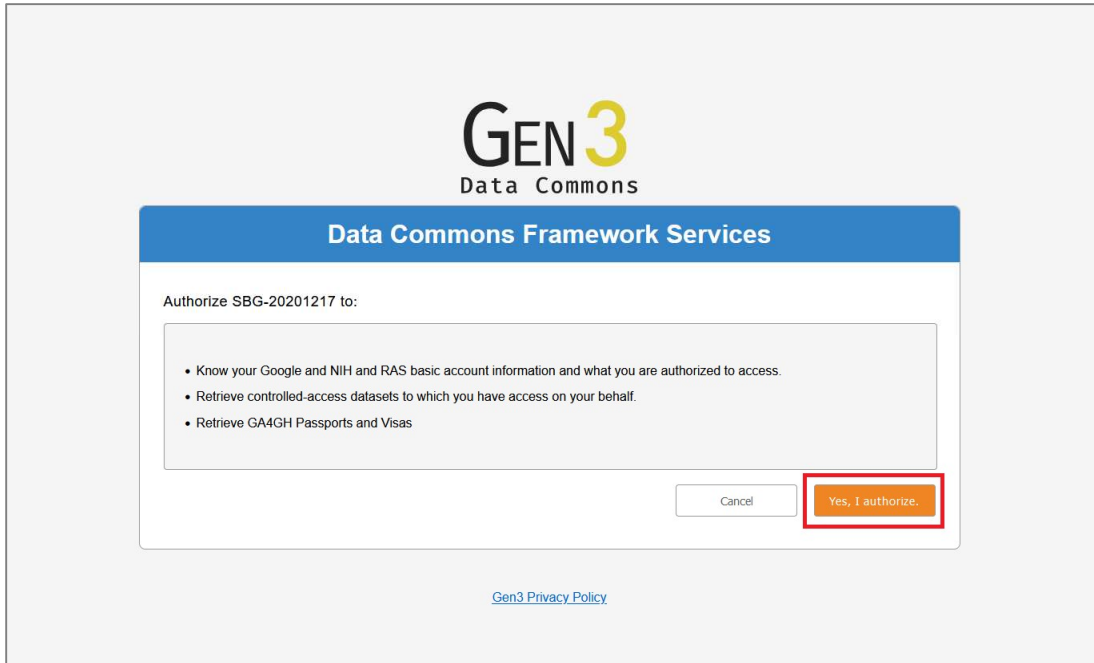
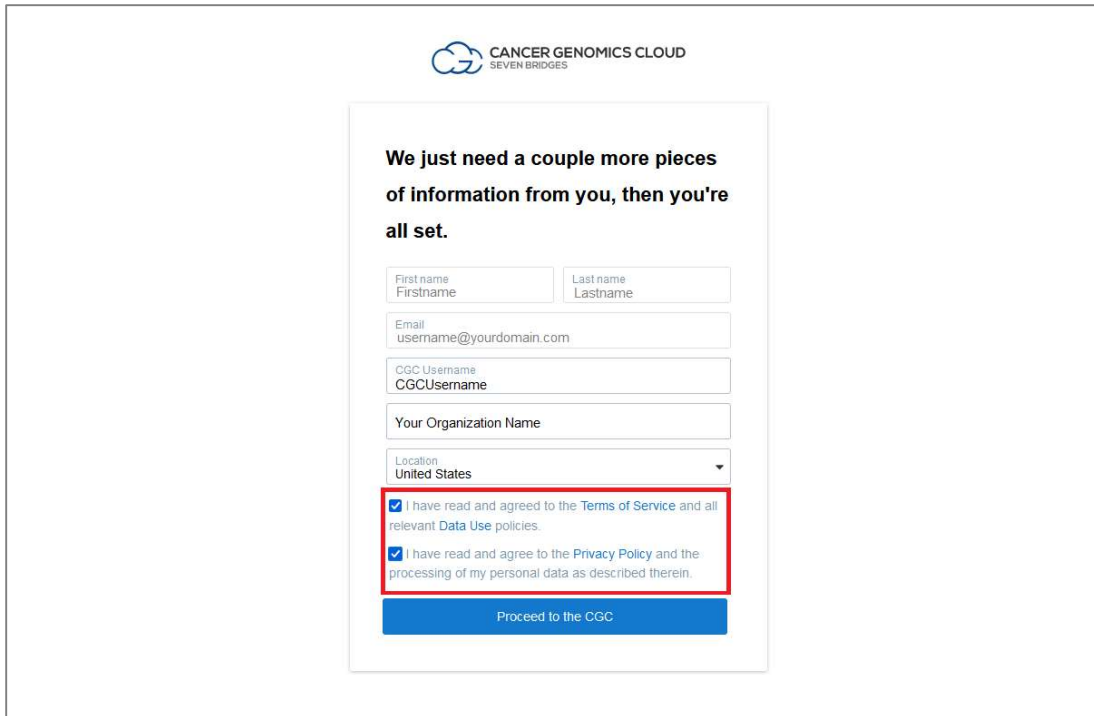


Figure C5: Gen3 DCFS authorization page with “Yes, I authorize” button highlighted in a red box

6. On the next page, confirm that the information listed for you is correct (if this page appears). If you agree to the Terms of Service, Data Use, and Privacy policies, click the two related checkboxes, and then click on “Proceed to the CGC” (Figure C6).



CANCER GENOMICS CLOUD
SEVEN BRIDGES

We just need a couple more pieces of information from you, then you're all set.

First name
Firstname

Last name
Lastname

Email
username@yourdomain.com

CGC Username
CGCUsername

Your Organization Name

Location
United States

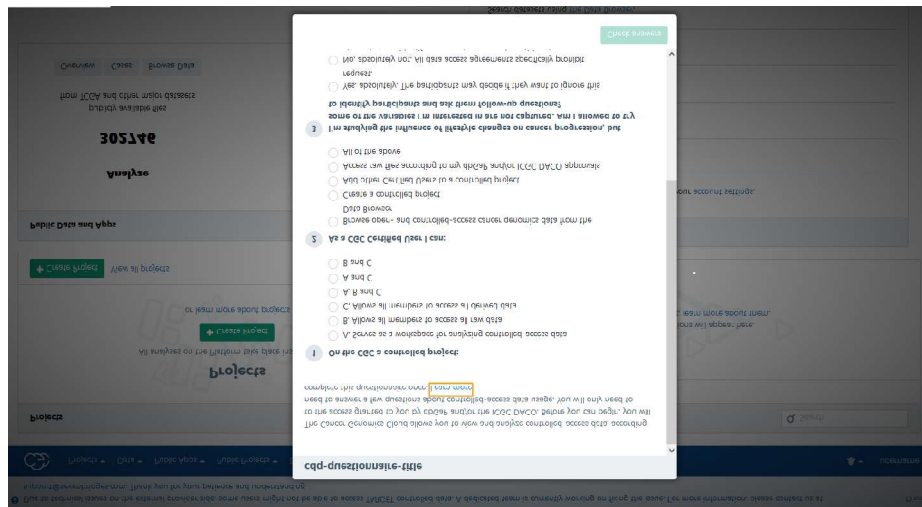
☒ I have read and agreed to the [Terms of Service](#) and all relevant [Data Use](#) policies.

☒ I have read and agree to the [Privacy Policy](#) and the processing of my personal data as described therein.

Proceed to the CGC

Figure C6: Confirmation of terms and policy for CGC registration highlighted in a red box

7. If the CGC questionnaire appears, complete it to continue (Figure C7).



CGC questionnaire

1. On the CGC a completed project:

2. As a CGC member I can:

3. I am a CGC member because:

4. I am a CGC member because:

5. I am a CGC member because:

6. I am a CGC member because:

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100. I am a CGC member because:

Figure C7: Screenshot of the CGC questionnaire

Creating a DRS Manifest from downloaded Study metadata

The Study Manifest file contains all metadata about a study available in the CCDI Hub. This file can be used to create a Data Repository Service (DRS) file that includes identifiers and metadata that allows users to locate and access specific files or data sets within a data repository for use on the CGC. To convert the Study Manifest file into a DRS manifest, please follow these steps:

1. Go to ccdi.cancer.gov/explore

2. Select "Molecular Characterization Initiative" under Study Name, or choose "phs002790" under Study dbGaP Accession.
3. Download the Study Manifest (contains metadata for the study).
4. Find the tabs ending with "_file" (look for FASTQ, JSON, etc.).
5. Create a DRS Manifest for Cloud Analysis
 - Extract and rename the column headers: file_name (name); dcf_indexd_guid (drs_uri); study_id (study_code); sample.sample_id/participant_id (case_id).
6. Add drs://nci-crdc.datacommons.io/ to all the beginning of all rows in the drs_uri column. Go to <https://www.cancergenomicscloud.org/>:
7. In CGC, go to your Project. Under Files, find Add Files and GA4GH Data Repository Service (DRS).
8. Upload your manifest.
9. On the next page, select the "From a manifest file" tab and drag and drop your manifest onto the page. Go through the next steps agreeing to the use of the system and ingest the manifest.
10. After being redirected back to the file page for the project, it will begin to load the files into the Files section of the CGC project.

Downloading files from the CGC

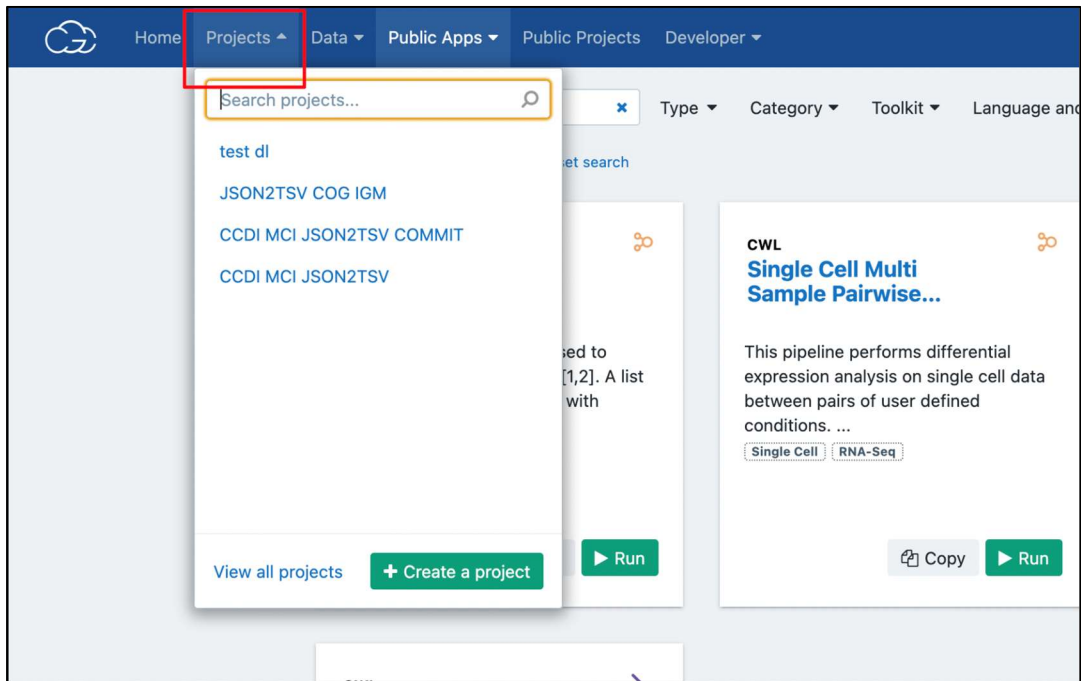
Data exported to CGC from CCDI Explore or produced from either CGC Public Apps or from data analysis in the CGC Data Studio will be output to Files tab. Please note, if the user intent is to download large and/or many files indexed in CCDI Explore, it is highly recommended that users utilize the Gen3 client with file DCF GUIDs as outlined in Appendix B: NCI Data Commons Framework Services (DCFS): Controlled Data Access Instructions.

There are several methods to download files from the CGC:

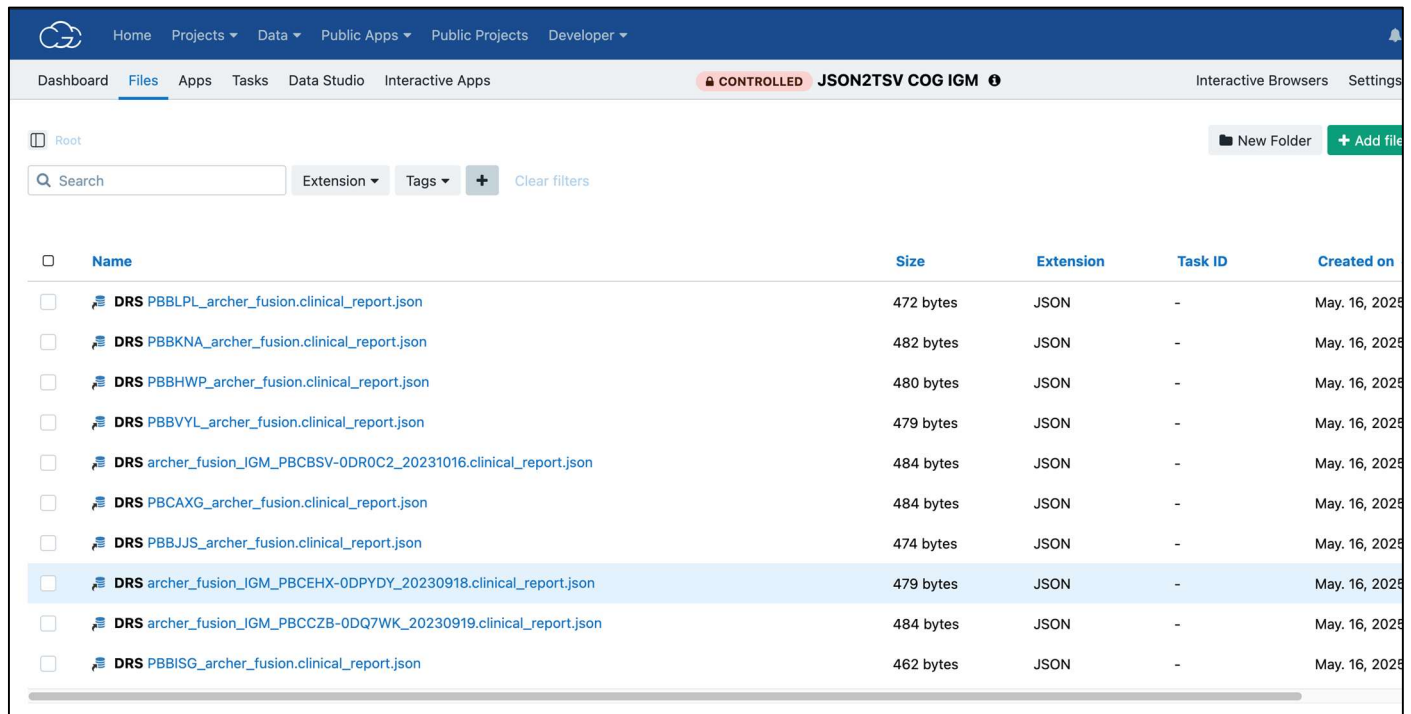
1. CGC Web Interface
2. Seven Bridges Command Line Interface (CLI)
 - a. Recommended method for downloading large and/or many files
3. Using Download Links

1. Download Files Using the CGC Web Interface

1. Navigate to the Project: Select the project (workspace) where your data is stored.

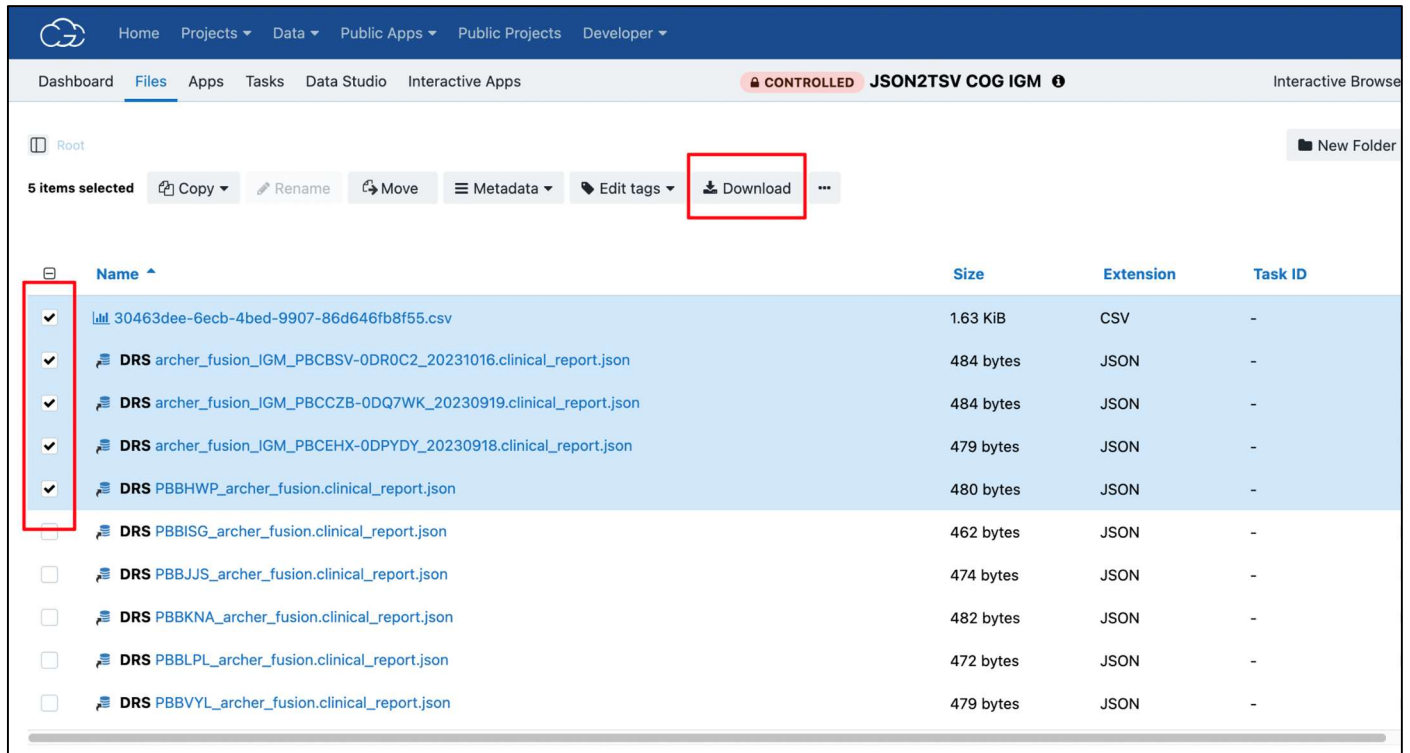


2. Open the Files Tab: Click on the “Files” tab within the project workspace.
3. Locate Your File(s): Browse or search for the specific files or folders you want to download.



4. Download Files:
 - Select one or more files using the checkboxes.

- Click on the Download icon (cloud with a downward arrow).
- If prompted, confirm the download location or process.



Note: The CGC has limitations on the number of files that can be downloaded at once.

2. Download Files Using Seven Bridges CLI

This method is suitable for downloading large datasets or automating workflows.

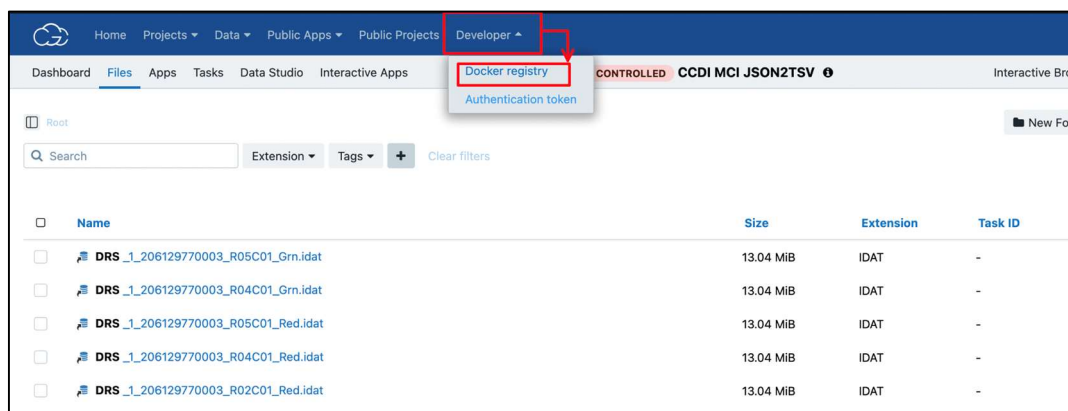
Prerequisites:

- Install the Seven Bridges CLI
 - Installation Guide: <https://docs.cancergenomicscloud.org/docs/command-line-interface>
- Authenticate the CLI with your CGC account
 - Enter following command in terminal: `sb configure`
 - User will then be prompted to enter in the following information:

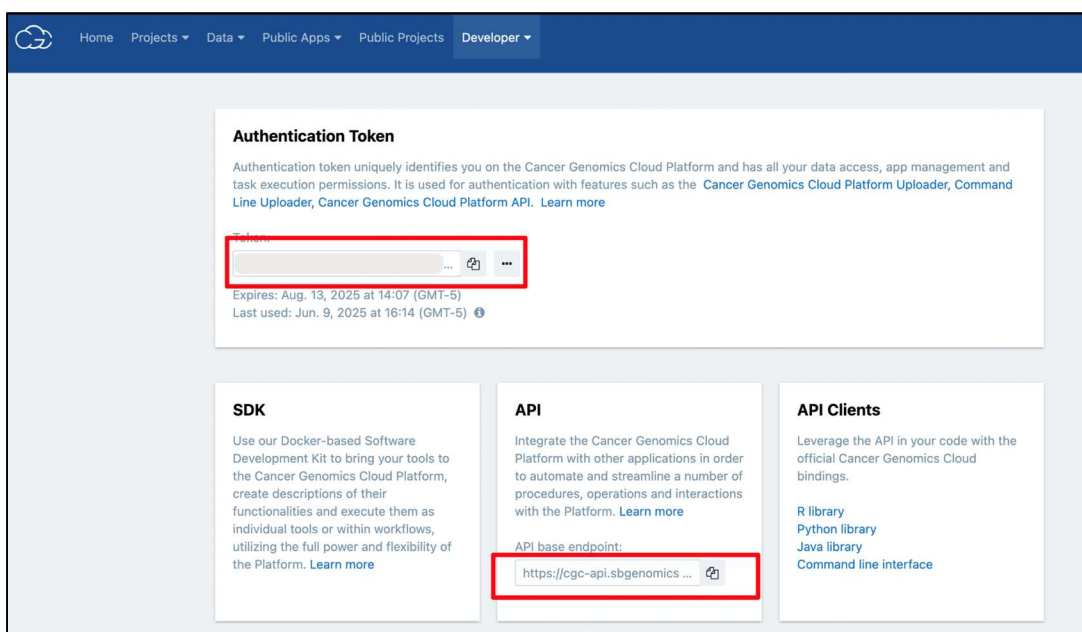
```
USER$ sb configure
Seven Bridges API endpoint [None]:
Seven Bridges authorization token [None]:
Seven Bridges advance access (true/false) [None]: false
```

- API Endpoint

- Authorization token
- Advance access True/False: Enter false unless advance access applies to your endpoint
- The API Endpoint and Authorization token can be found from the Developer > Authentication token page in the CGC Dashboard:

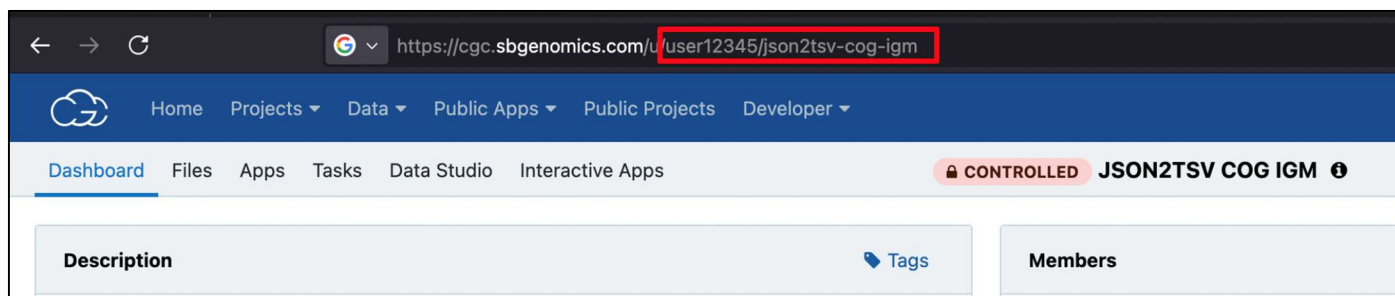


- And then copied from Authentication token page:



Example Commands

For following commands, replace project-name with actual name of your project, can be found from project URL in browser as “username/project_name. For example:



In example above, project name is “user12345/json2tsv-cog-igm”

List files in a project

`sb files list --project project-name`

Lists file_id, file_name and project location, can pipe to a file to see all files and ids in one location:

`sb files list --project project-name > project_files.txt`

Use returned file_ids to download specific files with CLI.

Download a file

`sb download --file <file_id_value> ... [--destination <destination_value>] [--overwrite] [flags]`

Download multiple files

`sb download --file <file_id_1> --file <file_id_2> [--overwrite]`

Display help information

`sb help` or `sb help [command]`, e.g. `sb help files` for seeing help menu for `sb files` command

For more information, please see the following relevant links at the CGC Documentation site:

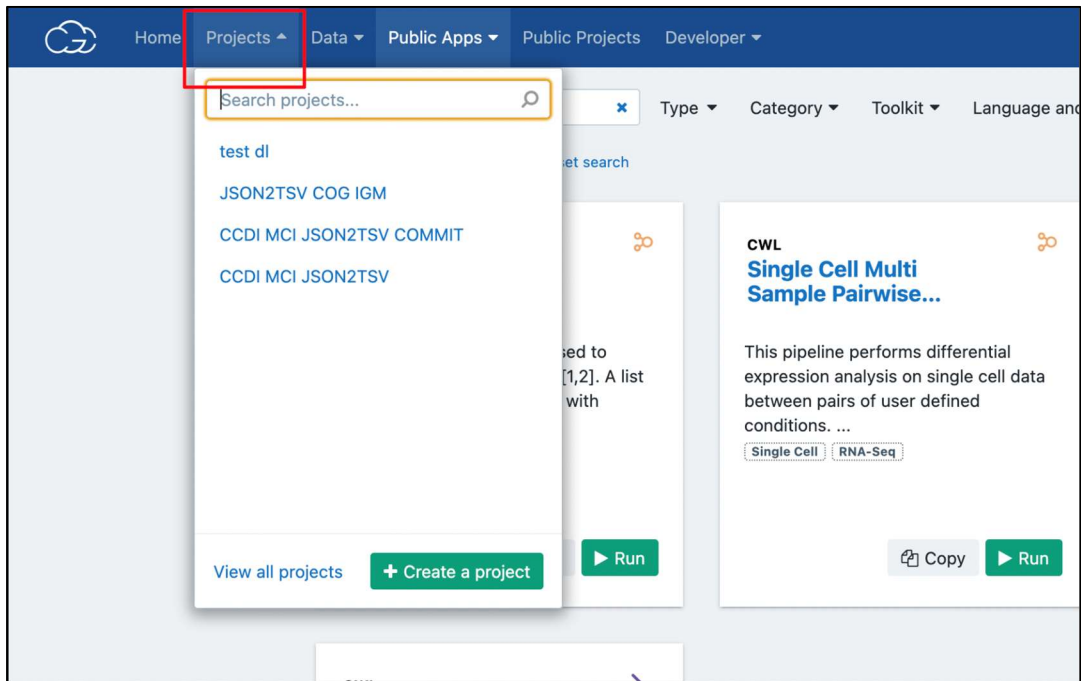
<https://docs.cancergenomicscloud.org/docs/command-line-interface>

<https://docs.cancergenomicscloud.org/docs/upload-and-download-files>

3. Download Files Using CGC Generated Download Links

Generated Download Links enable users to download files using curl, wget, aria2 or other preferred third-party download tools. Please note: there is a limit to the number of download links that can be generated at given time for file downloads, this method is not recommended for bulk download of files from the CGC. For more information, please see CGC Documentation site regarding Download Links: <https://docs.cancergenomicscloud.org/docs/download-results#download-files-using-direct-download-links>

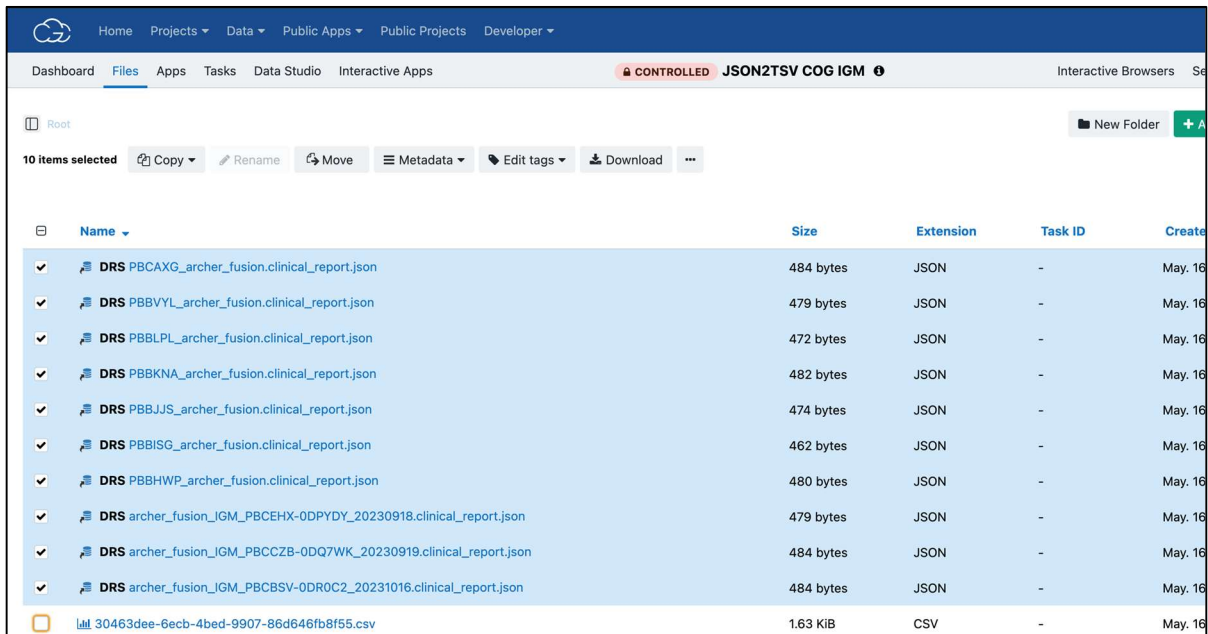
1. Navigate to the Project: Select the project (workspace) where your data is stored.



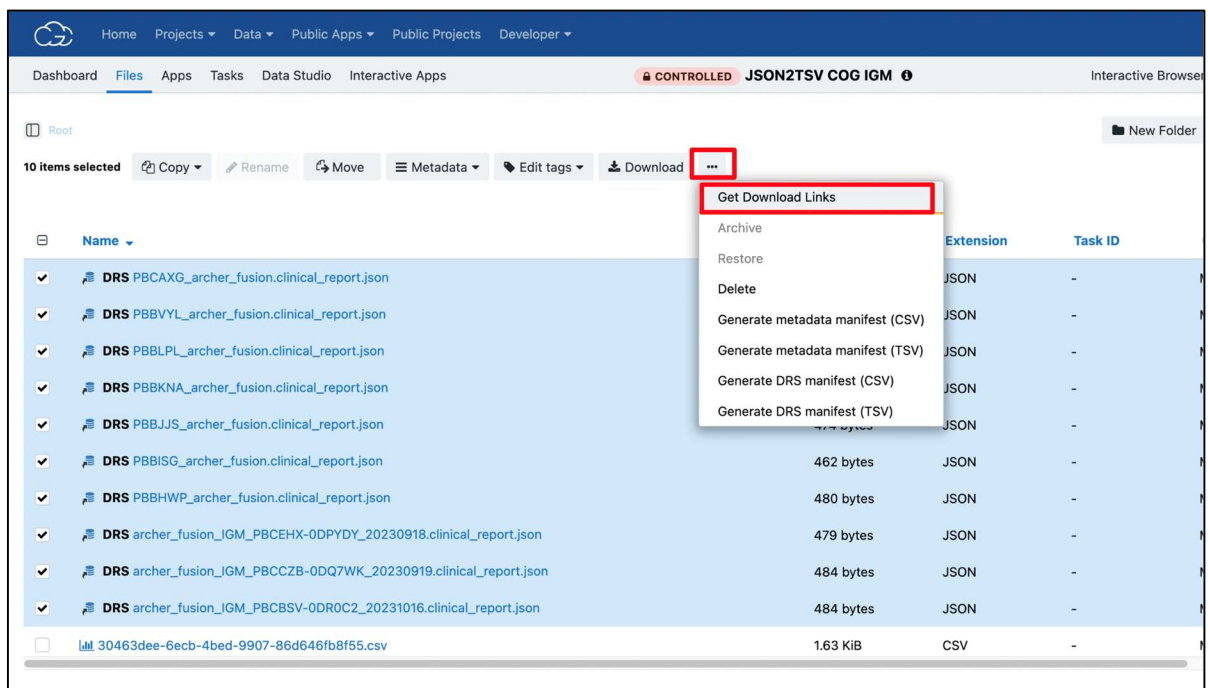
2. Open the Files Tab: Click on the “Files” tab within the project workspace.
3. Locate Your File(s): Browse or search for the specific files or folders you want to download.

Dashboard Files Apps Tasks Data Studio Interactive Apps CONTROLLED JSON2TSV COG IGM Interactive Browsers Settings					
Root Search Extension Tags + Clear filters New Folder Add file					
☐	Name	Size	Extension	Task ID	Created on
☐	DRS PBBPLP_archer_fusion.clinical_report.json	472 bytes	JSON	-	May. 16, 2025
☐	DRS PBBKNA_archer_fusion.clinical_report.json	482 bytes	JSON	-	May. 16, 2025
☐	DRS PBBHWP_archer_fusion.clinical_report.json	480 bytes	JSON	-	May. 16, 2025
☐	DRS PBBVYL_archer_fusion.clinical_report.json	479 bytes	JSON	-	May. 16, 2025
☐	DRS archer_fusion_IGM_PBCBSV-0DR0C2_20231016.clinical_report.json	484 bytes	JSON	-	May. 16, 2025
☐	DRS PBCAXG_archer_fusion.clinical_report.json	484 bytes	JSON	-	May. 16, 2025
☐	DRS PBBJJS_archer_fusion.clinical_report.json	474 bytes	JSON	-	May. 16, 2025
☐	DRS archer_fusion_IGM_PBCHEX-0DPYDY_20230918.clinical_report.json	479 bytes	JSON	-	May. 16, 2025
☐	DRS archer_fusion_IGM_PBCZB-0DQ7WK_20230919.clinical_report.json	484 bytes	JSON	-	May. 16, 2025
☐	DRS PBBISG_archer_fusion.clinical_report.json	462 bytes	JSON	-	May. 16, 2025

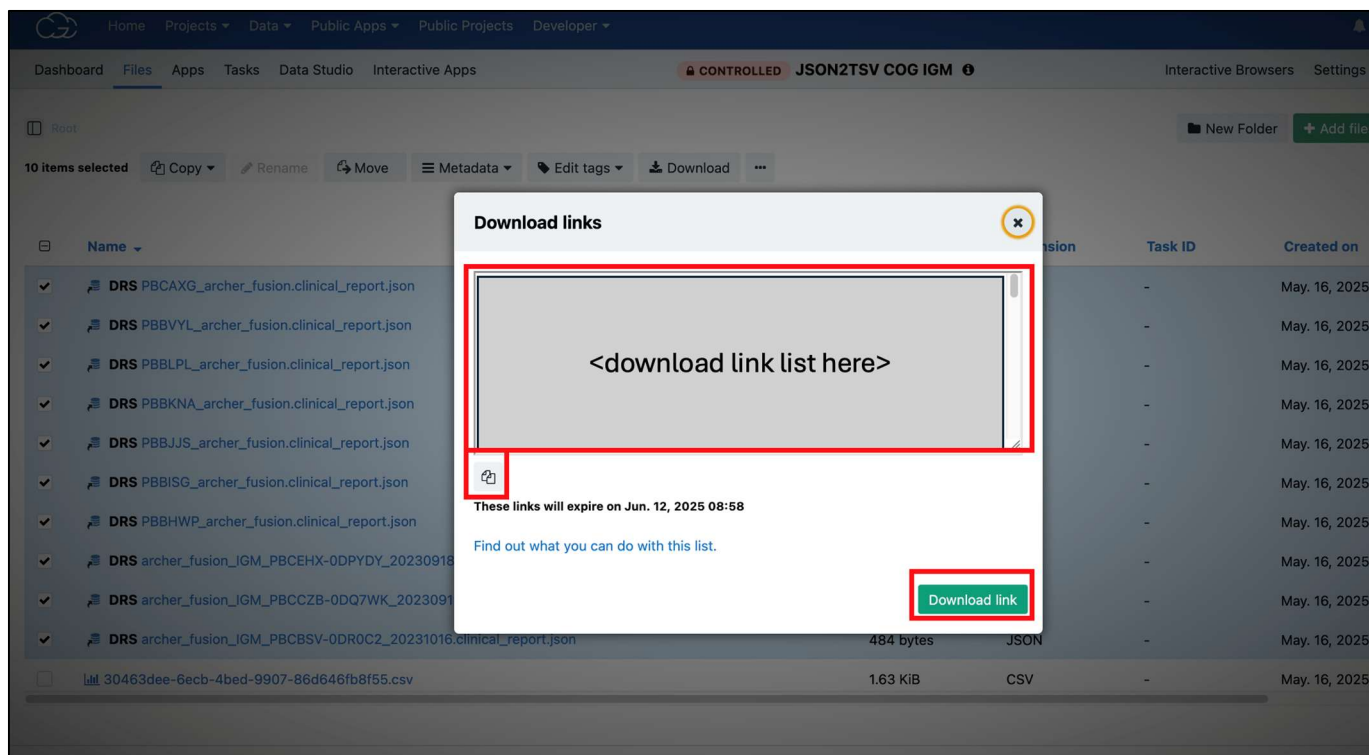
4. Select the files you would like to download. Note: There is a limit to the number of files that download links can be generated for at a given time, ~50 files.



- After files have been selected, select the ellipses button next to the “Download” button, and then from drop down menu select “Get Download Links”



- A popup with download links for selection of files is displayed; download links can be then copied from popup by clicking copy button or downloaded to a TXT file by clicking green “Download link” button



Please note: download links provide access to controlled access files and should not be shared with outside parties or unauthorized individuals. Download links expire 48 hours after generation.

7. Users can then download files with download links using preferred third party download app.

Download individual files with download link, example with curl:

```
curl -O "https://example-download-link-url"
```

Since output is streamed, user should specify -O flag ahead of link so that file download is saved to file with same name as source file. Depending on terminal settings, you may need to place parentheses around file download link to perform special character escaping.

Additional Resources on Working with Data at the CGC

- CGC Documentation: docs.cancergenomicscloud.org/docs
- Importing General Commons Data: <https://docs.cancergenomicscloud.org/docs/import-data-from-the-gc>
- Common Workflow Language Workflows and Apps: cgc.sbgenomics.com/public/apps
- Volumes: docs.cancergenomicscloud.org/docs/volumes-1
- Tool Editor Tutorial: docs.cancergenomicscloud.org/docs/tool-editor-tutorial
- About the Editor: docs.cancergenomicscloud.org/docs/about-the-editor